Bit-Level Verification of a Fixed-Point LIF Neuron and a 256-Neuron Spiking Network on a Low-Cost FPGA

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Abstract

Spiking neuron models are small enough to run on inexpensive FPGAs, but a hardware result is only useful for model-based experiments if one can state exactly how it differs from the model. We present a leaky integrate-and-fire (LIF) neuron and a 256-neuron network on a low-cost FPGA (Tang Nano 20K, Gowin GW2AR-18), and we test both against a Python reference model. The model uses signed Q8.8 fixed-point state, saturating additions and shift-based time constants, so no multiplier is needed. On the board, the single neuron matched the model at every step of 60,000 samples from three hand-picked parameter sets and of 2,023,200 samples from a sweep of 2,248 parameter sets. A time-multiplexed network with spike-driven synapses, controlled over a UART link, matched the model in every reply of a 682-frame command flow, and in the spike counts, the final-state snapshot and the first 512 steps of spikes of five excitation/inhibition (E/I) conditions. Mutation tests show that the checkers can fail. The full design uses 12% of the logic and 98% of the block RAM, and meets timing at 38.75 MHz after place and route, against a 27 MHz clock. In the model, the mean firing rate of the network falls monotonically as the inhibitory fraction rises, in every combination of the sweep. On the board, the current reached its upper and lower limits and the potential its lower limit, and every saturated step matched the model; the upper limit of the potential cannot change the output. The membrane potential of the network was compared only at the last step, so saturation inside the network was not checked. We also report the limits

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we found: a USB–UART link that lost bytes in long unpaced bursts, a command decoder without a time-out, and a spike trace that covers only the first 512 steps. The hardware is validated against the model only; we make no claim about biological validity.

1 Introduction

Networks of spiking neurons with excitatory and inhibitory populations are used to study how the balance between excitation and inhibition shapes neural activity. Such networks are small enough to run on inexpensive FPGAs, and many FPGA designs for spiking networks have been reported [1].

A hardware result is only useful for model-based experiments if one knows how it differs from the model it implements. Fixed-point hardware makes this question sharp: quantisation, saturation and the choice between arithmetic and logical shifts each change the numbers, and an error in any of them can leave the average behaviour almost intact while changing individual spikes. Validation by task accuracy, or by agreement with a simulator, can miss such errors, and some FPGA reports validate their designs this way [2, 3]. A stricter test asks that the hardware state after every step equals the state of a reference model, with zero differences, so that a single flipped bit makes the test fail.

Bit-accurate software models are also used as references for FPGA designs. An integer-LIF accelerator for image classification was checked against a bit-accurate software model on the full MNIST test set, while the hardware was checked for structural equivalence on a 1,000-sample subset, rather than step by step [4]. A systolic simulator built around updating only the neurons that spike is an example of a design aimed at speed and scale [5]. These designs target other boards and tasks, and their numbers are not comparable with ours.

This paper applies the strict test to a single neuron and to a recurrent network with excitatory and inhibitory sources, on a low-cost board. We make three contributions.

1. **An exact specification.** A fixed-point LIF neuron with saturating additions and arithmetic shifts, and a network input rule that sums the synaptic contributions exactly and saturates once. Because the sum is exact, the result does not depend on the order in which the sources are added.

2. **A verification method.** The model is the specification. The method combines simulation, mutation tests and board comparison. For the

single neuron, the board comparison is step by step (membrane potential and spike) and covers 2,083,200 samples. For the network, the board comparison covers a 682-frame command flow, and for five E/I conditions the spike counts, the final-state snapshot and the spikes of the first 512 steps. The membrane potential of the network is compared only at the last step.

3. **A 256-neuron network on a low-cost board.** A UART-controlled, time-multiplexed network, with its resource use and the limits we measured: the link lost bytes in long unpaced bursts, the decoder has no time-out, and the spike trace covers 512 steps.

We also include an E/I experiment in the model, in which the fraction of inhibitory sources is varied, and we run five of its conditions on the board. Its purpose is to show that the hardware and the model agree while the E/I balance changes. It does not support conclusions about neural tissue.

The paper is organised as follows. Section 2 defines the model, Section 3 the hardware, and Section 4 the verification method. Section 5 gives the board results and the E/I demonstration, Section 6 the resource use, Section 7 the limitations, and Section 8 concludes.

2 Model

2.1 Single neuron

Each step updates a membrane potential V and a synaptic current I:

$$I[t+1] = I[t] - (I[t] \gg k_s) + x[t], \tag{1}$$

$$V[t+1] = V[t] + ((V_{\text{rest}} - V[t]) \gg k_m) + I[t+1], \tag{2}$$

where $x[t]$ is the input of the step and every addition saturates (Section 2.2). We write $s[t]$ for the spike flag of step t. If the neuron is not refractory and $V[t+1] \geq \theta$, it emits a spike, $s[t+1] = 1$, V is set to V_{reset}, and the neuron is refractory for *refrac* steps. While it is refractory, V is held at V_{reset} and I keeps integrating. The parameters are θ, V_{reset}, V_{rest}, k_m and k_s (0–15), and *refrac* (0–7). A right shift by k divides by 2^k, so the time constants are $\Delta t/\tau = 2^{-k}$.

2.2 Number format

V and I are signed 16-bit numbers in Q8.8 format (raw value = real value $\times 256$). Three rules fix the arithmetic exactly:

- Every addition saturates to $[-32768, 32767]$ after each addition, from left to right. Nothing wraps.
- Right shifts are arithmetic (they round toward $-\infty$).
- $V_{\text{rest}} - V$ is computed with more than 16 bits before the shift, because the difference of two 16-bit values may need 17 bits.

A wrap would turn a strong excitatory input into a strong inhibitory one, and a logical shift would turn a decaying negative potential into a large positive one. The mutation tests in Section 4 include the logical shift. They do not cover every wrap or saturation variant, which is a limitation (Section 7). The reference is a short Python function (`model/fixed.py`); the hardware must match it exactly.

2.3 Network

All neurons share one parameter set. Let $s_j[t]$ be the spike flag of neuron j at step t. The input of neuron i at step t is

$$x_i[t] = \text{sat}\left(\sum_j \left(W_{ji} \ll w_{\text{shift}}\right) s_j[t-1] \; + \; \text{ext}_i \right), \tag{3}$$

where W_{ji} is a signed 8-bit weight from neuron j to neuron i (self-connections are allowed, and a zero weight means no connection), $w_{\text{shift}} = 2$ is a global scale, and ext_i is a constant external input. The sum over sources is exact, the external input is added to it, and the total saturates once. This differs from the single-neuron rule on purpose. The hardware visits the sources in a fixed order, and an exact sum followed by a single saturation gives the same result for every order. A spike at step t reaches the inputs at step $t + 1$.

3 Hardware

3.1 Board and link

The target is a Tang Nano 20K with a Gowin GW2AR-18 device and a 27 MHz clock. A UART link at 115,200 baud connects it to a PC. The toolchain is open source (yosys, nextpnr-himbaechel, openFPGALoader). Frames have the form `[0xA5][cmd][len][data][checksum]`, with an additive checksum. The decoder on the FPGA rejects a frame with a bad checksum or an out-of-range value, and counts accepted and rejected frames. It

refuses every command while it is busy sending a reply, copying a payload or running the network.

3.2 Single neuron

One neuron step takes one clock cycle: the update is a combinational circuit, and the state is held in registers. Each step writes {spike, V} (17 bits) to a record memory of 1,024 steps in block RAM. The host reads the records back in batches. Each run starts from rest ($V = 0$, $I = 0$, no refractory time), the same initial condition as the model.

3.3 Network

The network of $N = 256$ neurons uses one copy of the neuron circuit, shared in time: one step takes N neuron updates followed by the synapse phase. All memories except the spike flags have one write port and one synchronous read port, so the synthesis tool maps them to block RAM. The spike flags are small and are read combinationally, so they are kept in distributed memory. Written as arrays, the whole state would become flip-flops; an earlier array version needed about 45,000 cells at $N = 256$, which does not fit the chip. Synaptic input is spike driven: when neuron j spikes, its weight row is added to an accumulator for the next step. Two accumulators alternate, one read while the other is written. Both phases are pipelined with a one-cycle read latency. The read order keeps consecutive reads on different neurons, so the pipeline never reads a value that is still being written.

The board does not keep the full state of every step. At $N = 256$ and 1,024 steps, the full state would need 4,352 Kbit ($256 \times 1{,}024 \times 17$ bits), and the block RAM holds about 828 Kbit. The board therefore keeps, per neuron, a 16-bit spike count and a 17-bit snapshot of the last step (33 bits in total), and the spike flags of every neuron for the first 512 steps of a run (1 bit per neuron and step, 128 Kbit), returned seven steps per frame. The membrane potential is therefore compared only at the last step, which is the main limit of the network verification (Section 7).

4 Verification

The model is the specification, and a difference of one bit is a failure. The verification has several layers.

Simulation. The single neuron was checked with cocotb and Icarus Verilog against the model: 10,110 combinational cases and 15 traces of 500 steps each. The network core was checked against the model for $N = 16, 64, 256$ over 3,200, 12,800 and 51,200 records, and the full design was checked at the byte level through the UART pins for the same sizes (for $N = 256$, 682 frames).

Mutation tests. A checker that has never failed does not show that it can fail. Each checker was therefore run on deliberately broken copies of the design. Each of the roughly 15 hand-written mutants was caught by its checker. Examples: replacing the arithmetic shift with a logical shift (single neuron); a weight shift of 3, a missing last target, no spike delivery, and a write to the wrong accumulator (network core); swapped input bytes, an off-by-one range check and a wrong read-modify-write timing (full design); a changed bit order, a step address off by one, a limit of 513 instead of 512 steps and a limit of eight instead of seven steps per frame (spike trace); and a status read that ignores the busy signal (busy test). This is not a complete mutation analysis, and it does not show that every possible error is caught. An early byte-level testbench of the single neuron exposed a real design error: a second run was sent while the previous reply was still being sent, which overwrote the record memory. The busy rule of Section 3 is the fix.

Board, single neuron. The host writes the parameters, confirms them with a status read, runs the neuron, reads every step's V and spike back and compares them with the model.

Board, network. The host sends the full command flow of 682 frames (70,168 bytes in, 1,892 bytes out) and compares every reply byte with the model's prediction. The flow contains deliberately invalid frames, which must be refused without changing any state. The E/I conditions are compared in the same way: spike counts, last-step snapshot and the spikes of the first 512 steps against the full model.

No netlist simulation. The block-RAM simulation models that come with the installed toolchain are empty, so a netlist simulation cannot run. We therefore use the board comparison as the functional evidence for the synthesised design. It covers the cases we ran, not all cases.

Table 1: Single neuron: board against model. Values are real numbers; the Q8.8 raw value is the real value times 256.

Set	θ	V_{rest}	V_{reset}	k_m	k_s	refrac	Samples	Mismatches
A	2.0	-0.25	0	4	3	2	20,000	0
B	0.5	0.25	-0.5	0	0	0	20,000	0
C	2.0	0	-0.5	15	15	7	20,000	0

4.1 AI-assisted development

This work was developed with Claude Code, an AI coding assistant. The assistant wrote and edited code, tests and text under the direction of the author. The author chose the design, the verification method and the claims, and checked the results. The author is responsible for every claim in this paper.

5 Results

5.1 Single neuron on the board

Table 1 shows three parameter sets with 20 constant input currents each (from -0.1 to 0.375) and 1,000 steps per input. All 60,000 samples matched the model, and the error counter of the board stayed at zero. Figure 1 shows the first 100 steps of parameter set A, and Figure 2 the f–I points of the same run against the model.

The fixed-point and floating-point versions of the same neuron agree in shape, with a largest difference of 0.027 spikes per step (`docs/fi_curve.png`). That comparison uses the defaults of `lif.Params` ($\theta = 1.0$, $V_{\text{reset}} = 0$, $V_{\text{rest}} = 0$, $k_m = 4$, $k_s = 2$, $\mathit{refrac} = 2$), input currents from 0 to 0.1 and 2,000 steps. It does not use the parameter sets of Table 1.

5.2 Parameter sweep

Table 2 covers every legal value of k_m, k_s and *refrac*, and 200 random parameter sets. Each set ran three constant inputs (-0.2, 0.1, 0.4) for 300 steps. Across both sweeps, 2,023,200 samples matched with no mismatch, no failed read and no refused request. Sets without spikes (24 grid, 5 random) are included on purpose: a silent neuron has to match the model too.

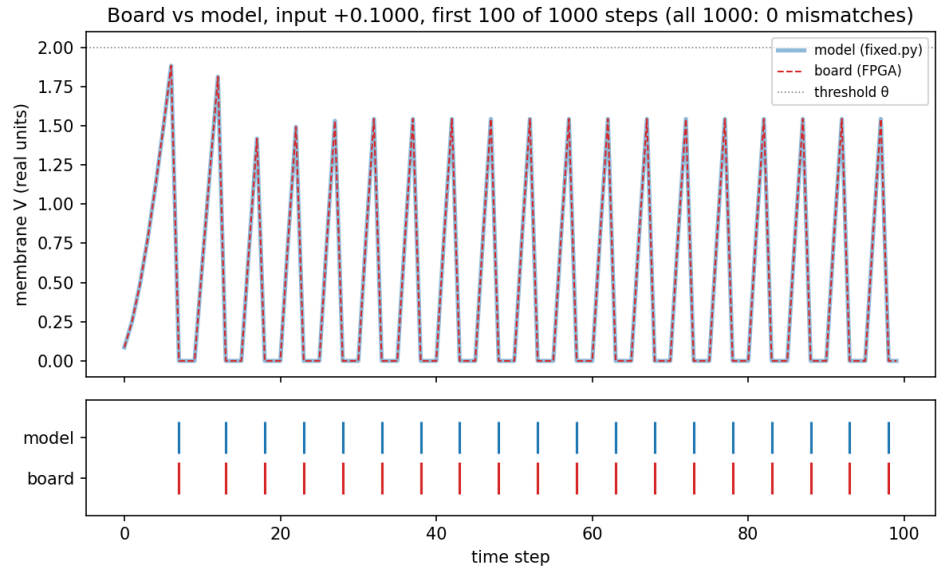


Figure 1: Membrane potential and spikes of parameter set A, board against model, first 100 of 1,000 steps. The potential is the recorded value after the step, so at a spike it shows the reset value rather than a crossing of θ.

5.3 Saturation on the board

The single-neuron update has four saturating additions: the decay of I (i_1), the input added to I (i_2), the leak of V (v_1) and the current added to V (v_2). The first two cannot saturate. The decay $I - (I \gg k_s)$ lies between I and zero, and the leak $V + ((V_{\text{rest}} - V) \gg k_m)$ lies between V and V_{rest}, so both stay in the 16-bit range. The upper limit of v_2 cannot change the output either. A value clamped there is 32,767 $\geq \theta$, so the neuron spikes, or it is refractory and the potential is discarded; in both cases the stored potential is V_{reset}, as it would be without the clamp. The other three limits change the stored state or the spike decision, and they need a test.

Table 3 lists the runs that reached them. The board does not report I, so a clamp of I is seen only through the potential it produces. Clamp counts come from a traced copy of the model (`host/clamp_census.py`, output in `results/clamp_census.log`); the copy is asserted to equal `fixed.step` at every step. The saturation test (`host/saturation_board_test.py`) uses the largest shifts, $k_m = k_s = 15$, and constant inputs at the ends of the range, so that I reaches both limits and V reaches its lower limit on the

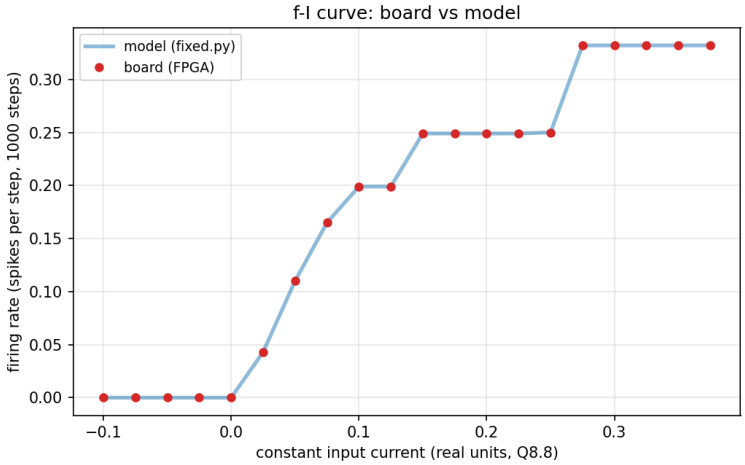


Figure 2: f–I points of parameter set A, board and model, for the 20 constant inputs. The input is in real units (the Q8.8 raw value divided by 256); the rate is spikes per time step over 1,000 steps, not Hz. Each step reflects an integer inter-spike interval: for a neuron that fires with a fixed interval T the rate is $1/T$, and the refractory period bounds the rate from above. The dead zone of the arithmetic shift only affects the onset near threshold.

board. No run reached the upper limit of v_2.

5.4 Network on the board

The 682-frame flow at $N = 256$ gave 1,892 reply bytes, all identical to the model, in 44.4 s (run of 2026-11-06, with the current `board_network_test.py`; log in `results/`). Five E/I conditions then ran on the board, each with 745 frames: $(f_{\text{inh}}, w_{\text{inh}}, w_{\max}, e_{\max}) = (0.2, 1, 10, 30)$, $(0.5, 1, 10, 30)$, $(0.5, 0.5, 10, 30)$, $(0.5, 2, 10, 30)$ and $(0.3, 1.5, 20, 10)$. For each, the spike counts, the last-step snapshot and the spikes of the first 512 steps matched the full model. Clamp events inside the network were not counted. Its membrane potential is compared only at the last step (Section 7), so a clamp there would not be seen; the largest input magnitudes were 1,084 raw in the command flow and 252 to 622 raw in the E/I runs. The saturation of the single neuron is reported in Section 5.3.

Table 2: Parameter sweep on the board. Grid: $\theta = 1.0$, $V_{\text{reset}} = -0.25$ and $V_{\text{rest}} = 0.125$ fixed, k_m, k_s and *refrac* over all legal values. Random: seed 20261103, each parameter drawn uniformly over the ranges $\theta \in [64, 1024]$, $V_{\text{reset}} \in [-512, 256]$, $V_{\text{rest}} \in [-512, 512]$, $k_m, k_s \in [0, 15]$ and *refrac* $\in [0, 7]$ (voltages as Q8.8 raw values).

Sweep	Sets	Samples	Mismatches	Failed reads	Sets with spikes
Grid	2,048	1,843,200	0	0	2,024
Random	200	180,000	0	0	195

Table 3: Runs that reached a saturation limit, and their comparison with the model. Limits: I and V are the current and potential updates, with upper or lower limit.

Runs	Limits reached	Runs with a clamp	Samples	Mis
Single neuron, set C (Table 1)	I upper, V lower	14 of 20	20,000	
Sweep, grid (Table 2)	V lower	1,400 of 2,048 sets	1,843,200	
Sweep, random (Table 2)	V lower	130 of 200 sets	180,000	
Saturation test, sets S1 and S2	I upper and lower, V lower	10 of 10 runs	1,000	

5.5 E/I demonstration (model)

The demonstration uses random networks of 256 neurons with a connection probability of $8/N$, a fraction f_{inh} of inhibitory sources, and excitatory weights drawn from $[0, w_{\max}]$ with $w_{\max} = 10$. Inhibitory weights are $-w_{\text{inh}}$ times a weight of the same range, with $w_{\text{inh}} = 1$ unless stated. Each neuron has a constant external input drawn from $[0, e_{\max}]$ (Q8.8 raw) with $e_{\max} = 30$. Each run has 600 steps, and the first 100 are discarded. Table 4 gives the mean firing rate in spikes per step and per neuron, over 10 random networks. The spread is the minimum and maximum over the networks. The rate falls as f_{inh} rises, and the share of silent neurons rises. The full sweep covers 24 combinations of $w_{\max}$, w_{inh} and $e_{\max}$, each with seven values of f_{inh} and 10 networks (1,680 runs). In all 24 combinations the mean rate falls monotonically with f_{inh}, and it does so in every individual network. The five board conditions use the same generator and the first seed (20261104), so each board condition is one of the networks of this sweep. We had to lower $w_{\max}$ from 60 to 10: with 60, the network was either silent or at the refractory limit of 1/3 spikes per step, and the external input no longer mattered. At $f_{\text{inh}} = 0$, on average 78% of neurons fire above 0.3 spikes per

Table 4: Mean firing rate against f_{inh} ($w_{\max} = 10$, $e_{\max} = 30$, $w_{\text{inh}} = 1$), averaged over 10 random networks, with the minimum–maximum over networks in brackets, and the share of silent neurons. Source: `build/ei_full_sweep.csv`, from `host/ei_full_sweep.py`, the generator also used for the board runs. Model only. Rates are per time step, not in Hz.

f_{inh}	Mean rate (min–max)	Silent neurons
0.0	0.314 (0.301–0.320)	0.00
0.1	0.290 (0.273–0.296)	0.01
0.2	0.251 (0.243–0.259)	0.03
0.3	0.210 (0.197–0.221)	0.07
0.4	0.176 (0.163–0.189)	0.13
0.5	0.143 (0.135–0.158)	0.20
0.6	0.121 (0.106–0.137)	0.26

step, close to the refractory limit of 1/3 (65% to 84% across the networks). The decrease at the low-inhibition end is therefore partly a ceiling effect. The useful range of the perturbation is about 0.2 to 0.6.

5.6 Link limits

The USB–UART link on the board lost bytes in long unpaced bursts: a single write of 96 or 128 bytes returned only 66 bytes in a loopback test, while 16-byte chunks paced to the line rate all came back. We did not isolate the cause; it may lie in the bridge chip or in the host driver. The host therefore sends in 16-byte chunks at line speed, and waits for the full reply of a frame before it sends the next one. A run takes no reply, so the host waits for a time computed from the number of steps. This is a heuristic bound that we did not verify against the board's actual run time (Section 7).

6 Resources and scaling

Table 5 gives the place-and-route result of the full design at $N = 256$. The single-neuron logic (`lif_core` alone, an early yosys run without the state memory) needed about 274 LUTs. The yosys cell counts of the network core, 1,382, 1,343 and 1,457 at $N = 16, 64, 256$, include I/O buffers and the RAM16SDP4 cells that the tool uses for small memories, so they are not comparable across N. Block RAM is what limits the design: the weight

Table 5: Full design at $N = 256$ (nextpnr-himbaechel, GW2AR-18). Source: `fpga/build/make_network_top_2026-11-04.log`, a complete rebuild (`make -B TOP=network_top`); the maximum clock is the critical-path report after routing.

Resource	Used	Available
LUT4	2,536	20,736 (12%)
Flip-flops	1,315	15,552 (8%)
Block RAM	45	46 (98%)
Maximum clock, after routing	38.75 MHz	27 MHz target, met

matrix alone is $N^2 \times 8$ bits (512 Kbit), and the spike trace takes eight more blocks. With 45 of 46 blocks used, we cannot add a feature without first freeing memory. For $N = 512$ the weight matrix would be 2 Mbit, more than the roughly 828 Kbit of block RAM on the chip. This is arithmetic from the N^2 weight matrix, not a built design.

7 Limitations

- **Biology.** The hardware is validated against the model only. The E/I demonstration uses synthetic random networks, weights that are not calibrated against data, and rates per time step, not in Hz. We draw no conclusion about neural tissue.
- **Saturation.** The three clamps that can change the output were reached on the board, and every saturated step matched the model (Section 5.3). The upper clamp of the potential was not reached in any traced run, and it cannot change the output. Clamps inside the network were not counted, and the network's potential was compared only at the last step.
- **Stimulus.** On the board, the input is a constant per neuron, and the sweep uses three constant inputs for 300 steps. Time-varying input was not tested on the board.
- **Membrane potential in the network.** The board compares spikes for the first 512 steps, and V only at the last step. An intermediate V would need a run of that length from rest for each step. A run of more than 512 steps keeps only the counts and the last snapshot after step 512.

- **Reference model.** The Python model is the only reference. It was written with AI assistance (Section 4), and a misreading of the specification would be shared by the model and the hardware, so the board comparison cannot detect it. The unit tests in `model/test_fixed.py` check the arithmetic and the update rules on selected cases; they are not an independent implementation. Floating-point simulators are not bit-exact and cannot serve as that reference.

- **Mutation analysis.** About 15 hand-written mutants were run, and no full kill matrix was produced. The cocotb test of the single neuron exercises one mutant, the logical shift.

- **No netlist simulation.** The block-RAM models of the installed toolchain are empty. The synthesis tool prints range warnings for some memory mappings. The board result is our evidence that these do not matter, but it is a test of the cases we ran, not a proof.

- **Parameter space.** All neurons share one parameter set, there is no noise source, and the grid sweep fixes θ, V_{reset} and V_{rest} (the random sweep covers them).

- **Protocol.** The additive checksum does not detect swapped bytes. The decoder has no time-out, so a corrupted length field can misalign the frame stream. Adding one would need an RTL change, new simulation tests and a new board run, and we have not made that change. Some builds accept `READ_RECORD` and `READ_TRACE` without sending a reply and count them as successful, so the host must not send them. The board has no reset command, so its counters persist until the bitstream is reloaded, and the command-flow test must start from a freshly loaded board.

- **Timing.** The wait after a run is computed from the number of steps as a worst-case heuristic. We did not verify it against the time the board actually needs.

- **Design choices.** The protocol, the weight scale w_{shift} and the choice of the E/I test network are our own choices. They were not tuned or compared with alternatives.

- **Connectivity.** The nominal connection probability is $8/N$, but zero weights are excluded, so the effective density is lower. For $f_{\text{inh}} = 0.2$ the networks have on average 2.8% nonzero entries of the 256×256 matrix (range 2.7% to 2.9%), against a nominal 3.1%.

- **Samples.** Each E/I data point uses 10 random networks and one external-input sample per network. The board ran five conditions of the 24 combinations of the sweep.

8 Conclusion

We implemented a fixed-point LIF neuron and a 256-neuron network on a low-cost FPGA, and compared both with a Python reference model. The single neuron matched the model step by step on the board, over 2,083,200 samples. The network matched the model in its command replies, its spike counts and final-state snapshot, and in the first 512 steps of spikes for five E/I conditions. The three clamps that can change the output were reached on the board and matched the model step by step (Section 5.3). The membrane potential of the network was compared only at the last step, and clamps inside the network were not counted. The E/I results are model results for synthetic networks and are not a biological claim.

Availability

The code, RTL, testbenches, host scripts and raw results are available in the public repository `https://github.com/KiraIch/fpga-lif-snn`, branch `m3-network`. The raw results (sweep CSVs, board logs and place-and-route logs) are in the `results/` directory. A release of the code is archived on Zenodo with the DOI 10.5281/zenodo.24011633.

Declarations

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Author contributions. The work was carried out independently by the author, ChiaAn Lin.
Ethics. Not applicable: no human or animal data were used.

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